

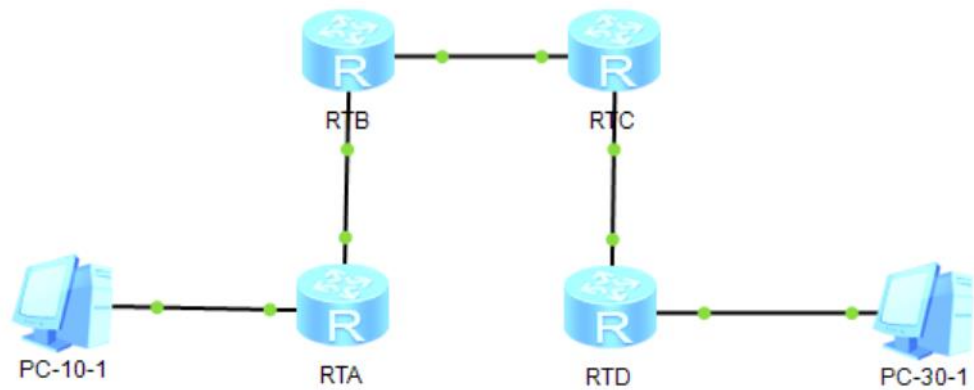
Lab-5 Basic RIP Configuration and Analysis

学号：110221981 姓名：刘方坤晗 行政班级：07022204

一、Lab-5-1: Basic RIPv1 Configuration

Q5-1.1~5-1.8:

(1) 创建的拓补图如下：



(2) RTA 的 IP 路由表如下：

```
[RTA]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
    Destinations : 12          Routes : 12
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	Static	60	0	RD	172.16.101.2	GigabitEthernet0/0/0
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
172.16.101.0/24	Direct	0	0	D	172.16.101.1	GigabitEthernet0/0/0
172.16.101.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
172.16.101.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
192.168.10.0/24	Direct	0	0	D	192.168.10.1	GigabitEthernet0/0/2
192.168.10.1/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
192.168.10.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
192.168.30.0/24	Static	60	0	RD	172.16.101.2	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

(3) RTB 的 IP 路由表如下：

```
[RTB]display IP routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 13          Routes : 13

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
0.0.0.0/0           Static  60   0          RD   172.16.102.2     GigabitEthernet
0/0/1
127.0.0.0/8         Direct  0    0          D    127.0.0.1        InLoopBack0
127.0.0.1/32        Direct  0    0          D    127.0.0.1        InLoopBack0
127.255.255.255/32   Direct  0    0          D    127.0.0.1        InLoopBack0
172.16.101.0/24      Direct  0    0          D    172.16.101.2     GigabitEthernet
0/0/0
172.16.101.2/32     Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/0
172.16.101.255/32    Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/0
172.16.102.0/24      Direct  0    0          D    172.16.102.1     GigabitEthernet
0/0/1
172.16.102.1/32     Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/1
172.16.102.255/32    Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/1
192.168.10.0/24      Static  60   0          RD   172.16.101.1     GigabitEthernet
0/0/0
192.168.30.0/24      Static  60   0          RD   172.16.102.2     GigabitEthernet
0/0/1
255.255.255.255/32   Direct  0    0          D    127.0.0.1        InLoopBack0
```

(4) RTC 的 IP 路由表如下:

```
[RTC]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 13          Routes : 13

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface
-----
0.0.0.0/0           Static  60   0          RD   172.16.103.1     GigabitEthernet
0/0/0
127.0.0.0/8         Direct  0    0          D    127.0.0.1        InLoopBack0
127.0.0.1/32        Direct  0    0          D    127.0.0.1        InLoopBack0
127.255.255.255/32   Direct  0    0          D    127.0.0.1        InLoopBack0
172.16.102.0/24      Direct  0    0          D    172.16.102.2     GigabitEthernet
0/0/1
172.16.102.2/32     Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/1
172.16.102.255/32    Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/1
172.16.103.0/24      Direct  0    0          D    172.16.103.2     GigabitEthernet
0/0/0
172.16.103.2/32     Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/0
172.16.103.255/32    Direct  0    0          D    127.0.0.1        GigabitEthernet
0/0/0
192.168.10.0/24      Static  60   0          RD   172.16.102.1     GigabitEthernet
0/0/1
192.168.30.0/24      Static  60   0          RD   172.16.103.1     GigabitEthernet
0/0/0
255.255.255.255/32   Direct  0    0          D    127.0.0.1        InLoopBack0
```

(5) RTD 的 IP 路由表如下:

```
[RTD]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 14          Routes : 14

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
0.0.0.0/0           Static   60    0              RD   172.16.103.2      GigabitEthernet
127.0.0.0/8         Direct   0      0              D    127.0.0.1         InLoopBack0
127.0.0.1/32        Direct   0      0              D    127.0.0.1         InLoopBack0
127.255.255.255/32   Direct   0      0              D    127.0.0.1         InLoopBack0
172.16.103.0/24      Direct   0      0              D    172.16.103.1      GigabitEthernet
172.16.103.1/32      Direct   0      0              D    127.0.0.1         GigabitEthernet
172.16.103.255/32    Direct   0      0              D    127.0.0.1         GigabitEthernet
192.168.30.0/24      Direct   0      0              D    192.168.30.1      GigabitEthernet
192.168.30.1/32      Direct   0      0              D    127.0.0.1         GigabitEthernet
192.168.30.255/32    Direct   0      0              D    127.0.0.1         GigabitEthernet
192.168.200.0/24     Direct   0      0              D    192.168.200.100   LoopBack0
192.168.200.100/32   Direct   0      0              D    127.0.0.1         LoopBack0
192.168.200.255/32   Direct   0      0              D    127.0.0.1         LoopBack0
255.255.255.255/32   Direct   0      0              D    127.0.0.1         InLoopBack0
```

(6) PC-10-1 和 PC-30-1 可以互相 ping 通，结果截图如下：

```
PC>ping 192.168.30.11

Ping 192.168.30.11: 32 data bytes, Press Ctrl_C to break
From 192.168.30.11: bytes=32 seq=1 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=2 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.30.11: bytes=32 seq=4 ttl=124 time=16 ms
From 192.168.30.11: bytes=32 seq=5 ttl=124 time=15 ms

--- 192.168.30.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 15/24/31 ms
```

```
PC>ping 192.168.10.11

Ping 192.168.10.11: 32 data bytes, Press Ctrl_C to break
From 192.168.10.11: bytes=32 seq=1 ttl=124 time=32 ms
From 192.168.10.11: bytes=32 seq=2 ttl=124 time=31 ms
From 192.168.10.11: bytes=32 seq=3 ttl=124 time=31 ms
From 192.168.10.11: bytes=32 seq=4 ttl=124 time=31 ms
From 192.168.10.11: bytes=32 seq=5 ttl=124 time=16 ms

--- 192.168.10.11 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 16/28/32 ms
```

(7) PC-10-1 和 loopback 0 可以互相 ping 通，结果截图如下：

```
PC>ping 192.168.200.100

Ping 192.168.200.100: 32 data bytes, Press Ctrl_C to break
From 192.168.200.100: bytes=32 seq=1 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=2 ttl=252 time=31 ms
From 192.168.200.100: bytes=32 seq=3 ttl=252 time=32 ms
From 192.168.200.100: bytes=32 seq=4 ttl=252 time=15 ms
From 192.168.200.100: bytes=32 seq=5 ttl=252 time=32 ms

--- 192.168.200.100 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 15/28/32 ms
```

(8) PC-10-1 上 “tracert 192.168.30.11” 命令的结果截图如下：

```
PC>tracert 192.168.30.11

tracert to 192.168.30.11, 8 hops max
(ICMP), press Ctrl+C to stop
 1  192.168.10.1      16 ms   16 ms   15 ms
 2  172.16.101.2     31 ms   16 ms   <1 ms
 3  172.16.102.2     31 ms   16 ms   31 ms
 4  172.16.103.1     16 ms   31 ms   16 ms
 5  192.168.30.11    31 ms   16 ms   31 ms
```

Q5-1.9~5-1.14

- (9) 一共有两种 RIPv1 报文类型，分别是 Request（路由器启动时主动请求邻居的路由表）和 Response（周期性或触发更新的路由表广播）
- (10) RIPv1 默认大约每隔 30s 发送一次路由更新
- (11) RIPv1 使用的是 UDP 协议，源端口和目的端口均为 520
- (12) RIPv1 更新报文的目的 IP 地址为 255.255.255.255，是广播地址
- (13) 单条 RIPv1 更新报文最多包含 25 条路由，每条路由由包含 AFI 标识地址类型、目标网络 IP 地址、Metric 跳数等字段，截图如下：

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.16.102.1	255.255.255.255	RIPv1	86	Response
2	32.031000	172.16.102.1	255.255.255.255	RIPv1	86	Response
3	62.031000	172.16.102.1	255.255.255.255	RIPv1	86	Response
4	93.047000	172.16.102.1	255.255.255.255	RIPv1	86	Response

>	Frame 1: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface	0000	f
>	Ethernet II, Src: HuaweiTechno_ce:08:79 (00:e0:fc:ce:08:79), Dst: Broadcast (f	0010	e
>	Internet Protocol Version 4, Src: 172.16.102.1, Dst: 255.255.255.255	0020	f
>	User Datagram Protocol, Src Port: 520, Dst Port: 520	0030	e
>	Routing Information Protocol	0040	e
>		0050	e

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- > Ethernet II, Src: HuaweiTechno_0d:02:53 (00:e0:fc:0d:02:53), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- > Internet Protocol Version 4, Src: 172.16.103.1, Dst: 255.255.255.255
- > User Datagram Protocol, Src Port: 520, Dst Port: 520
- ✓ Routing Information Protocol
 - Command: Response (2)
 - Version: RIPv1 (1)
 - > IP Address: 172.16.103.0, Metric: 1
 - > IP Address: 192.168.30.0, Metric: 1
 - ✓ IP Address: 192.168.200.0, Metric: 1
 - Address Family: IP (2)
 - IP Address: 192.168.200.0
 - Metric: 1

0000	ff ff ff ff ff ff 00 e0 fc 0d 02 53 08 00 45 c0	S..E..
0010	00 5c 00 71 00 00 01 11 a5 4f ac 10 67 01 ff ff	..\.q....O.g...
0020	ff ff 02 08 02 08 00 48 6b cf 02 01 00 00 00 02	Hk.....
0030	00 00 ac 10 67 00 00 00 00 00 00 00 00 00 00	g.....
0040	00 01 00 02 00 00 c0 a8 1e 00 00 00 00 00 00	
0050	00 00 00 00 00 01 00 02 00 00 c0 a8 c8 00 00 00	

(14) RIPv1 更新报文格式:

Command (1)	Version (RIPv1=1)	必须为 0
AFI (IPv4=2)		必须为 0
目标 IP 地址 (4)		
子网掩码 (4)		
下一跳 (4)		
跳数 (4)		

Command 为 1 时是 Request 类型；为 2 时是 Response 类型；
跳数为 1-15,16 表示不可达